



BIOLOGICAL SCIENCE (Genetics)

Gregor Mendel: "Father of Genetics"

- **Trait** – a specific characteristic, such as petal color or eye color that varies from one individual to another.
- Usually traits have contrasting characters for example red petal vs. white petal or tall vs. short
- **Hybrid** – offspring of crosses between parents with different traits
- **Genes** – sections of DNA that determine traits
- **Allele** – different forms of a gene
- **Segregation** – allele pairs separate or segregate during gamete formation, and randomly unite at fertilization

Genotype	Phenotype
Combination of allele pairs	Physical manifestations of allele
a) Homozygous – genotype includes a pair of similar alleles (both dominant or recessive)	
b) Heterozygous – genotype includes a pair of different alleles (one dominant and one recessive)	

DNA

- Genetic material passed down from one generation to another.
- Composed of a phosphate group, deoxyribose (5-carbon sugar) and nitrogenous base.
- **4 Nitrogenous Base:** Adenine (A), Thymine (T), Guanine (G), Cytosine (C)
- **Base Pairing:** A – T; G – C

RNA

- Uracil is substituted for thymine, therefore the bases are A, U, G, C
- **Messenger RNA (mRNA)** – carries DNA message from nucleus and ribosomes

- **Transport RNA (tRNA)** – transports amino acids
- **Ribosomal RNA (rRNA)** – identification code of each ribosome

Transcription	Synthesis of mRNA while using the DNA as the template
Translation	Synthesis of the polypeptide chain within the ribosomes
Codons	Letter codes that stand for a certain amino acid found in the mRNA

PRINCIPLES OF GENETICS

• DOMINANCE

- ☐ Some alleles are dominant, while others are recessive
- ☐ An organism with a dominant allele for a particular form of a trait will always have that form
- ☐ An organism with a recessive form of a trait will only exhibit that form when the dominant allele is not present.

Example: The dominant allele for the trait seed color of a pea plant is yellow while the recessive one is gray. Now a pea plant with yellow allele and a gray one will always exhibit yellow seeds. A pea plant will exhibit gray seeds only if both of its alleles are gray.

Exemptions to Mendelian Genetics	
1. Incomplete dominance	One allele is not completely dominant over the other Example: White roses crossed with red roses result to pink roses.
2. Co-dominance	Both alleles contribute to the phenotype of organism Example: Type A and B are both dominant alleles, so the blood type becomes AB